

PHYSICS – BOARD QUESTION PAPER (FEBRUARY 2025)**Time: 3 Hours****Max. Marks: 70****General Instructions:**

1. Section A: Q1 contains 10 MCQs (1 mark each). Q2 contains 8 very short answers (1 mark each).
2. Section B: Q3 to Q14 contain 12 short answer questions (2 marks each). Attempt any 8.
3. Section C: Q15 to Q26 contain 12 questions (3 marks each). Attempt any 8.
4. Section D: Q27 to Q31 contain 5 long answer questions (4 marks each). Attempt any 3.
5. Use of log table is allowed. Calculator is not allowed.
6. Figures to the right indicate full marks.
7. For MCQs, only the first attempt will be considered.

Physical Constants:

- (i) $h = 6.63 \times 10^{-34} \text{ Js}$
- (ii) $c = 3 \times 10^8 \text{ m/s}$
- (iii) $\pi = 3.142$
- (iv) $g = 9.8 \text{ m/s}^2$
- (v) $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
- (vi) $L = 4\pi \times 10^{-7} \text{ Wb/A-m}$
- (vii) $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
- (viii) $1 \text{ atm} = 1.013 \times 10^5 \text{ N/m}^2$
- (ix) $R = 8.319 \text{ J/mol-K}$

SECTION A**Q1. Choose the correct alternatives:**

- i. The statement "If two systems are each in thermal equilibrium with a third system, they are also in thermal equilibrium with each other" refers to:

(A) Zeroth law (B) First law (C) Second law (D) Carnot's law

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ii. In Bernoulli's theorem, which quantity remains constant?

(A) Linear momentum (B) Angular momentum (C) Mass (D) Energy

iii. Which of the following is diathermanous?

(A) Wood (B) Iron (C) Glass (D) Water

iv. Electric potential V at distance r from a point charge is proportional to:

(A) r (B) $1/r$ (C) r^2 (D) $1/r^2$

v. Correct equation for internal resistance using potentiometer:

(A) $E/V = R(1 - l/l)$

(B) $I = R(1 - E/V)$

(C) $V/E = R(1 - l/l)$

(D) $r = R - 1$

vi. For same kinetic energy, the longest de-Broglie wavelength is for:

(A) Proton (B) Electron (C) Hydrogen atom (D) α -particle

vii. Gate giving high output when both inputs high:

(A) XOR (B) AND (C) NAND (D) NOR

viii. Power rating of ceiling fan at 2π rad/s:

(A) π W (B) 2π W (C) 3π W (D) 4π W

ix. A string of length 2 m vibrating in 2 loops. Distance between node & adjacent antinode:

(A) 0.5 m (B) 1.0 m (C) 1.5 m (D) 2.0 m

x. Transformer increases 220V to 880V. Primary = 1000 turns. Secondary turns:

(A) 1000 (B) 2000 (C) 3000 (D) 4000



Q2. Answer the following:

- i. Temperature at which surface tension of a liquid becomes zero.
- ii. Define self-inductance.
- iii. Work done by external uniform magnetic field on moving charge.
- iv. Define thermodynamic system.
- v. Name of B when $H = 0$ in hysteresis loop.
- vi. Formula for angle of banking.
- vii. Electric field near charged sheet ($\sigma = 8.85 \mu\text{C}/\text{m}^2$).
- viii. Kinetic energy of 1 litre ideal gas at STP.

SECTION B (Attempt any 8)

- Q3. Explain harmonics and overtones.
- Q4. Derive coefficient of viscosity using Newton's law.
- Q5. Magnetic induction in circular arc; find induction at center of loop.
- Q6. State and prove law of conservation of angular momentum.
- Q7. State any four advantages of LED.
- Q8. Calculate energy radiated in 1 minute by black body (area 200 cm^2) at 127°C .
- Q9. Two coils, self-inductances 60 mH each, coupling 0.75 . Find mutual inductance.
- Q10. $R = 32\Omega$, $X_L = 82\Omega$, $X_C = 40\Omega$. Find phase difference.
- Q11. Advantages of potentiometer over voltmeter.
- Q12. Draw neat labelled diagram of full-wave rectifier.
- Q13. Mass 0.8 kg in SHM, restoring force 0.4 N at 4 cm . Find k and period.
- Q14. 0.5 mole gas at 300 K expands isothermally from 2 L to 6 L . Work done.

SECTION C (Attempt any 8)

- Q15. Expression for period of bar magnet in SHM.
- Q16. Define mechanical, chemical, thermal equilibrium.
- Q17. Brewster's law and formula derivation.
- Q18. Derive radioactive decay law; define 1 becquerel.
- Q19. State and prove Kirchhoff's law of heat radiation.



Q20. End correction for open pipe & closed pipe.

Q21. Derive induced emf of rotating bar in perpendicular magnetic field.

Q22. Electron in second orbit completes 8 revolutions per 10 s. Find time period.

Q23. Flywheel mass 100 kg, radius 1.5 m, torque 2000 Nm. Work done in first 4 revolutions.

Q24. Convert galvanometer (50Ω , 2 mA) into (i) 0.5 A ammeter (ii) 10 V voltmeter.

Q25. Water drop diameter 0.6 mm. Calculate pressure inside drop.

Q26. Solenoid: length 1 m, diameter 5 cm, 1000 turns, 5A current. Find magnetic field at center.

SECTION D (Attempt any 3)

Q27. Explain ferromagnetism on basis of domain theory.

Q28. Expression for average power dissipated in LCR series circuit.

Q29. Interference vs diffraction. A double slit gives angular fringe separation 0.20° for $\lambda = 589 \text{ nm}$. Find separation in water (R.I. = 1.33).

Q30. Einstein's photoelectric equation; physical significance. $\lambda = 4000 \text{ \AA}$; calculate photon energy.

Q31. Four uses of Van de Graaff generator. Electric field = $2 \times 10^{11} \text{ V/m}$; plate area = 20 cm^2 . Find displacement current.

